

Chip Hong CHANG

professor

基本信息

姓名: Chip Hong CHANG 所属院校: 南洋理工大学
邮箱: echchang@ntu.edu.sg 地区: Singapore
年龄范围: 50岁以上 企业经历: 有

荣誉称号

★ IET Fellow

★ IEEE Fellow

★ AAIA Fellow

这些成就代表了在学术领域的最高认可，展现了卓越的研究能力和影响力。

学科领域

信息安全

电子与计算机工程

人工智能

集成电路设计与集成系统

网络空间安全

研究领域

深度学习与神经网络

网络安全理论与技术

数据隐私保护

嵌入式系统与硬件设计

安全协议与安全认证

加密与数据保护

计算机体系结构

电路与系统设计

低功耗电路与能量管理

密码学与密码分析

+3

个人简介

Dr. Chip Hong CHANG is a Professor at the School of Electrical & Electronic Engineering, Nanyang Technological University. He holds a PhD from Nanyang Technological University, Singapore. His research interests include hardware security, residue number systems, and more. He has had various employment

positions at NTU since 1990 and held concurrent appointments like Assistant Chair of Alumni. He has received numerous honors, such as the Microsoft Bounty Award, IEEE 2023 Outstanding Editorial Board Member Award, and is a Fellow of IEEE and AAIA. Contact: +65 67905873, echchang@ntu.edu.sg.

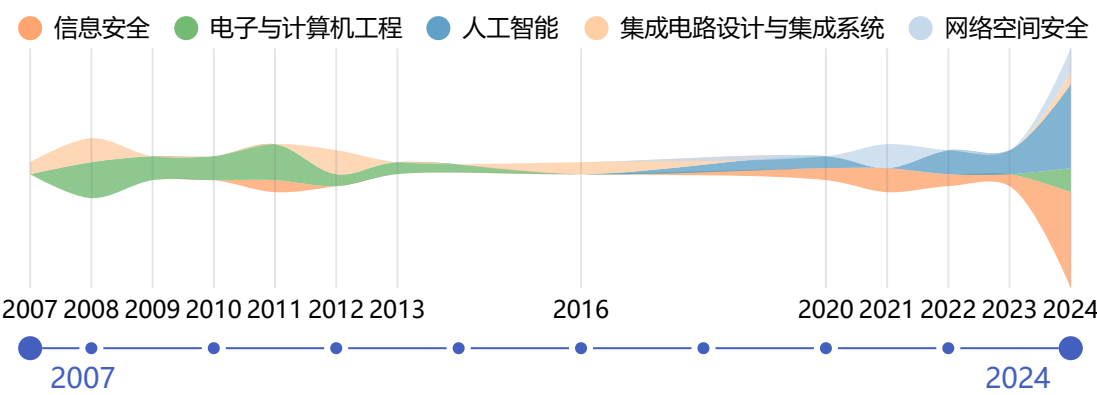
工作经历

Nanyang Technological University - Professor	2022-09-01 -
Nanyang Technological University - Associate Professor	2005-10-01 - 2022-08-31
Nanyang Technological University - Assistant Professor	1999-07-01 - 2005-09-30

教育经历

Nanyang Technological University - PhD	- 1998
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近年发表的论文



Live Demonstration: Man-in-the-Middle Attack on Edge Artificial Intelligence

2024 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS, ISCAS 2024

Hu, Bowen; He, Weiyang; Wang, Si; Liu, Wenye; Chang, Chip-Hong

被引用次数: 02024

SPFL: A Self-Purified Federated Learning Method Against Poisoning Attacks

IEEE TRANSACTIONS ON INFORMATION FORENSICS AND SECURITY

Liu, Zizhen; He, Weiyang; Chang, Chip-Hong; Ye, Jing; Li, Huawei; Li, Xiaowei

被引用次数: 02024

Steganographic Passport: An Owner and User Verifiable Credential for Deep Model IP Protection Without Retraining

IEEE/CVF Computer Vision and Pattern Recognition Conference

Qi Cui; Ruohan Meng; Chaohui Xu; Chip-Hong Chang

被引用次数: 0

2024

ASHES '24: Workshop on Attacks and Solutions in Hardware Security

ACM Conference on Computer and Communications Security

Lejla Batina; Chip Hong Chang; Ulrich Rührmair; Jakub Szefer

被引用次数: 0

2024

An Overview of Trustworthy AI: Advances in IP Protection, Privacy-Preserving Federated Learning, Security Verification, and GAI Safety Alignment

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Zheng, Yue; Chang, Chip-Hong; Huang, Shih-Hsu; Chen, Pin-Yu; Picek, Stjepan

被引用次数: 0

2024

Diffense: Defense Against Backdoor Attacks on Deep Neural Networks With Latent Diffusion

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Hu, Bowen; Chang, Chip-Hong

被引用次数: 0

2024

Steganographic Passport: An Owner and User Verifiable Credential for Deep Model IP Protection Without Retraining

2024 IEEE/CVF CONFERENCE ON COMPUTER VISION AND PATTERN RECOGNITION (CVPR)

Cui, Qi; Meng, Ruohan; Xu, Chaohui; Chang, Chip-Hong

被引用次数: 1

2024

LTE: Lightweight and Time-Efficient Hardware Encoder for Post-Quantum Scheme HQC

IEEE COMPUTER ARCHITECTURE LETTERS

Tu, Yazheng; He, Pengzhou; Chang, Chip-Hong; Xie, Jiafeng

被引用次数: 0

2024

Exploring Error Correction Circuits on RISC-V based Systems for Space Applications

2024 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS, ISCAS 2024

Koca, Nazim Altar; Chang, Chip Hong; Anh Tuan Do; Nambiar, Vishnu P.

被引用次数: 0

2024

Effect of Line Resistance of Passive Memristive Crossbars on Spiking Neural Network Performance

2024 IEEE 6TH INTERNATIONAL CONFERENCE ON AI CIRCUITS AND SYSTEMS, AICAS 2024

Lewden, Pierre; Vincent, Adrien F.; Tomas, Jean; Chang, Chip-Hong; Saighi, Sylvain

被引用次数: 0

2024

White-Box Adversarial Attacks on Deep Learning-Based Radio Frequency Fingerprint Identification

ICC 2023-IEEE INTERNATIONAL CONFERENCE ON COMMUNICATIONS

Ma, Jie; Zhang, Junqing; Shen, Guanxiong; Marshall, Alan; Chang, Chip-Hong

被引用次数: 1

2023

An Empirical Study of the Inherent Resistance of Knowledge Distillation Based Federated Learning to Targeted Poisoning Attacks

2023 IEEE 32ND ASIAN TEST SYMPOSIUM, ATS

He, Weiyang; Liu, Zizhen; Chang, Chip-Hong

被引用次数: 0

2023

Deep Texture-Depth-Based Attention for Face Recognition on IoT Devices

2022 IEEE ASIA PACIFIC CONFERENCE ON CIRCUITS AND SYSTEMS, APCCAS

Lin, Yuxin; Liu, Wenye; Chang, Chip-Hong

被引用次数: 0

2022

A Practical Man-in-the-Middle Attack on Deep Learning Edge Device by Sparse Light Strip Injection into Camera Data Lane

2022 IEEE 35TH INTERNATIONAL SYSTEM-ON-CHIP CONFERENCE (IEEE SOCC 2022)

Liu, Wenye; He, Weiyang; Hu, Bowen; Chang, Chip-Hong

被引用次数: 1

2022

An Overview of Hardware Security and Trust: Threats, Countermeasures, and Design Tools

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Hu, Wei; Chang, Chip-Hong; Sengupta, Anirban; Bhunia, Swarup; Kastner, Ryan; Li, Hai

被引用次数: 74

2021

A Modeling Attack Resistant Deception Technique for Securing Lightweight-PUF-Based Authentication

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Gu, Chongyan; Chang, Chip-Hong; Liu, Weiqiang; Yu, Shichao; Wang, Yale; O'Neill, Maire

被引用次数: 47

2021

Imperceptible Misclassification Attack on Deep Learning Accelerator by Glitch Injection

Design Automation Conference

Wenye Liu; Chip-Hong Chang; Fan Zhang; Xiaoxuan Lou

被引用次数: 0

2020

Special Issue on Convergence of Integrated Circuits and Complexity

JOURNAL OF CIRCUITS SYSTEMS AND COMPUTERS

Chang, Chip Hong; Kim, Tae Hyoung; Yu, Hao

被引用次数: 0

2016

An efficient channel clustering and flow rate allocation algorithm for non-uniform microfluidic cooling of 3D integrated circuits

INTEGRATION-THE VLSI JOURNAL

Qian, Hanhua; Chang, Chip-Hong; Yu, Hao

被引用次数: 18

2013

Area-Power Efficient Modulo $2n-1$ and Modulo $2n+1$ Multipliers for $\{2n-1, 2n, 2n+1\}$ Based RNS

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Muralidharan, Ramya; Chang, Chip-Hong

被引用次数: 29

2012

Special Section on the 2011 IEEE Custom Integrated Circuits Conference (CICC 2011)

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Chang, Chip-Hong; Luong, Howard; Pavan, Shanthi

被引用次数: 0

2012

Dynamical Systems Guided Design and Analysis of Silicon Oscillators for Central Pattern Generators

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Li, Fei; Basu, Arindam; Chang, Chip-Hong; Cohen, Avis H.

被引用次数: 8

2012

A VLSI Efficient Programmable Power-of-Two Scaler for $\{2n-1, 2n, 2n+1\}$ RNS

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Low, Jeremy Yung Shern; Chang, Chip-Hong

被引用次数: 16

2012

A Robust FSM Watermarking Scheme for IP Protection of Sequential Circuit Design

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Cui, Aijiao; Chang, Chip-Hong; Tahar, Sofiene; Abdel-Hamid, Amr T.

被引用次数: 84

2011

Simple, Fast, and Exact RNS Scaler for the Three-Moduli Set $\{2n-1, 2n, 2n+1\}$

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Chang, Chip-Hong; Low, Jeremy Yung Shern

被引用次数: 28

2011

Radix-8 Booth Encoded Modulo $2n-1$ Multipliers With Adaptive Delay for High Dynamic Range Residue Number System

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Muralidharan, Ramya; Chang, Chip-Hong

被引用次数: 25

2011

Sign-Extension Avoidance and Word-Length Optimization by Positive-Offset Representation for FIR Filter Design

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Huang, Ruimin; Chang, Chip-Hong; Faust, Mathias; Lotze, Niklas; Manoli, Yiannos

被引用次数: 5

2011

A Low Error and High Performance Multiplexer-Based Truncated Multiplier

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Chang, Chip-Hong; Satzoda, Ravi Kumar

被引用次数: 30

2010

On A New Common Subexpression Elimination Algorithm for Realizing Low-Complexity Higher Order Digital Filters

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Chang, Chip-Hong; Faust, Mathias

被引用次数: 17

2010

Synthesis-for-Testability Watermarking for Field Authentication of VLSI Intellectual Property

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Chang, Chip-Hong; Cui, Aijiao

被引用次数: 31

2010

High-Level Synthesis Algorithm for the Design of Reconfigurable Constant Multiplier

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Chen, Jiajia; Chang, Chip-Hong

被引用次数: 34

2009

A New Redundant Binary Booth Encoding for Fast $2n$ -Bit Multiplier Design

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

He, Yajuan; Chang, Chip-Hong

被引用次数: 31

2009

IP watermarking using incremental technology mapping at logic synthesis level

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Cui, Aijiao; Chang, Chip Hong; Tahar, Sofiene

被引用次数: 36

2008

A power-delay efficient hybrid carry-lookahead/carry-select based redundant binary to two's complement converter

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

He, Yajuan; Chang, Chip-Hong

被引用次数: 24

2008

Contention resolution - A new approach to versatile subexpressions sharing in multiple constant multiplications

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Xu, Fei; Chang, Chip-Hong; Jong, Ching-Chuen

被引用次数: 16

2008

Information theoretic approach to complexity reduction of FIR filter design

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Chang, Chip-Hong; Chen, Jiajia; Vinod, A. P.

被引用次数: 28

2008

Design of low-complexity FIR filters based on signed-powers-of-two coefficients with reusable common subexpressions

A Modeling Attack Resistant Deception Technique for Securing Lightweight-PUF-Based Authentication

TCAD

Chongyan Gu; Chip-Hong Chang; Weiqiang Liu; Shichao Yu; Yale Wang; Máire O' Neill

被引用次数: 0

An Overview of Hardware Security and Trust: Threats, Countermeasures, and Design Tools

TCAD

Wei Hu; Chip-Hong Chang; Anirban Sengupta; Swarup Bhunia; Ryan Kastner; Hai Li

被引用次数: 0

A Low-Power Hybrid RO PUF With Improved Thermal Stability for Lightweight Applications

TCAD

Yuan Cao; Le Zhang; Chip-Hong Chang; Shoushun Chen

被引用次数: 0

Scalable and Conflict-Free NTT Hardware Accelerator Design: Methodology, Proof, and Implementation

TCAD

Jianan Mu; Yi Ren; Wen Wang; Yizhong Hu; Shuai Chen; Chip-Hong Chang; Junfeng Fan; Jing Ye; Yuan Cao; Huawei Li; Xiaowei Li

被引用次数: 0

Optimizing Emerging Nonvolatile Memories for Dual-Mode Applications: Data Storage and Key Generator

TCAD

Le Zhang; Xuanyao Fong; Chip-Hong Chang; Zhi Hui Kong; Kaushik Roy

被引用次数: 0

New Hardware and Power Efficient Sporadic Logarithmic Shifters for DSP Applications

TCAD

Jiajia Chen; Chip-Hong Chang; Yujia Wang; Juan Zhao; Susanto Rahardja

被引用次数: 0

PUF-Based Mutual Authentication and Key Exchange Protocol for Peer-to-Peer IoT Applications

TDSC

Yue Zheng; Wenye Liu; Chongyan Gu; Chip-Hong Chang

被引用次数: 0

Highly Reliable Spin-Transfer Torque Magnetic RAM-Based Physical Unclonable Function With Multi-Response-Bits Per Cell

TIFS

Le Zhang; Xuanyao Fong; Chip-Hong Chang; Zhi Hui Kong; Kaushik Roy

被引用次数: 0

Static and Dynamic Obfuscations of Scan Data Against Scan-Based Side-Channel Attacks

TIFS

Aijiao Cui; Yanhui Luo; Chip-Hong Chang

被引用次数: 0

A DNN Fingerprint for Non-Repudiable Model Ownership Identification and Piracy Detection

TIFS

Yue Zheng; Si Wang; Chip-Hong Chang

被引用次数: 0

Reliable and Modeling Attack Resistant Authentication of Arbiter PUF in FPGA Implementation With Trinary Quadruple Response

TIFS

Siarhei S. Zalivaka; Alexander A. Ivaniuk; Chip-Hong Chang

被引用次数: 0

DW-AES: A Domain-Wall Nanowire-Based AES for High Throughput and Energy-Efficient Data Encryption in Non-Volatile Memory

TIFS

Yuhao Wang; Leibin Ni; Chip-Hong Chang; Hao Yu

被引用次数: 0

Ed-PUF: Event-Driven Physical Unclonable Function for Camera Authentication in Reactive Monitoring System

TIFS

Yue Zheng; Xiaojin Zhao; Takashi Sato; Yuan Cao; Chip-Hong Chang

被引用次数: 0

Stealthy and Robust Glitch Injection Attack on Deep Learning Accelerator for Target With Variational Viewpoint

TIFS

Wenye Liu; Chip-Hong Chang; Fan Zhang

被引用次数: 0

A PUF-Based Data-Device Hash for Tampered Image Detection and Source Camera Identification

TIFS

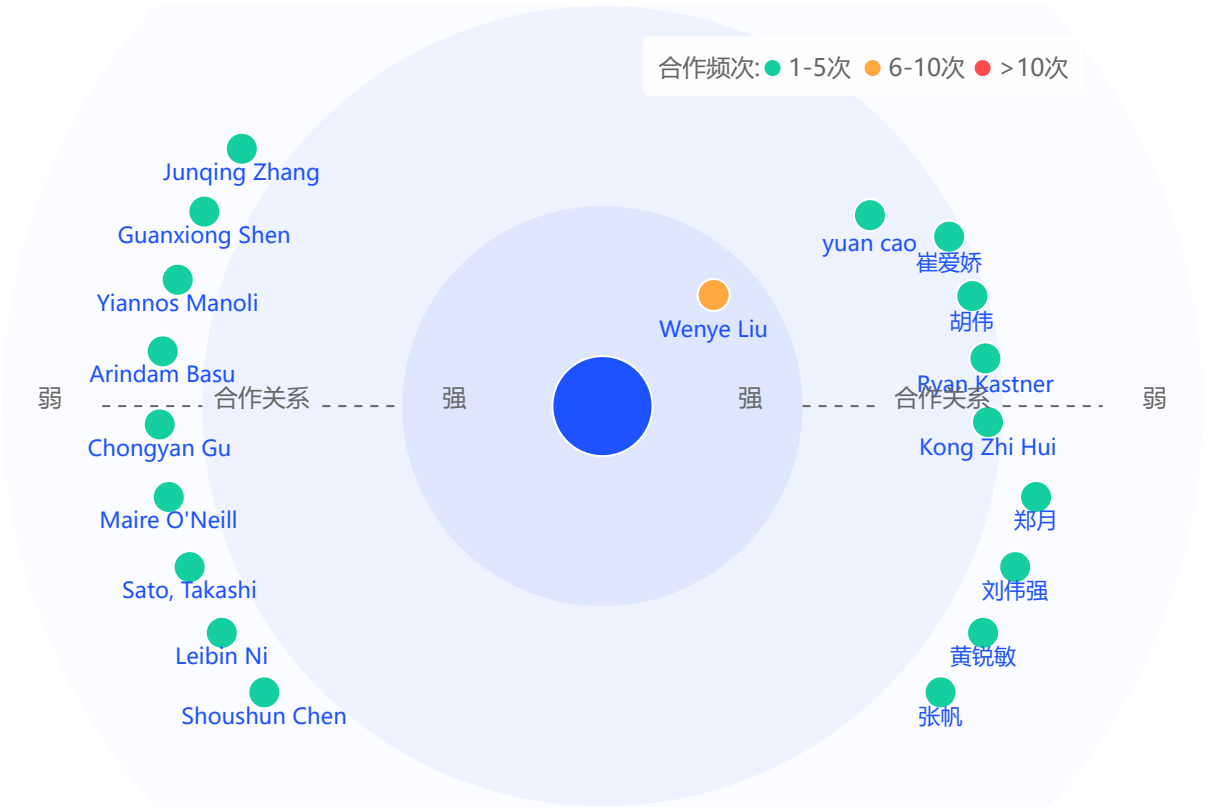
Yue Zheng; Yuan Cao; Chip-Hong Chang

被引用次数: 0

A non-iterative multiple residue digit error detection and correction algorithm in RRNS

TC
Thian Fatt Tay; Chip-Hong Chang
被引用次数: 0

学术合作



Wenye Liu	合作次数: 6
南洋理工大学	
yuan cao	合作次数: 3
南洋理工大学	
崔爱娇	合作次数: 2
哈尔滨工业大学	
胡伟	合作次数: 2
西北工业大学	
Ryan Kastner	合作次数: 2
加州大学-圣地亚哥	

还有 14 个合作者